



Mentigo: An Intelligent Agent for Mentoring Students in the Creative Problem Solving Process

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Abstract

Creative Problem-Solving (CPS) promotes creative and critical thinking while enhancing real-world problem-solving skills, making it essential for middle school education. However, providing personalized mentorship in CPS projects at scale is challenging due to resource constraints and diverse student needs. To address this, we developed Mentigo, an AI-driven mentor agent designed to guide middle school students through the CPS process. Using a dataset of real classroom interactions, we encoded CPS task stages, adaptive guidance strategies, and personalized feedback mechanisms to inform Mentigo's dynamic mentoring framework powered by large language models (LLMs). A comparative experiment with 12 students and evaluations from five expert educators demonstrated improved student engagement, creativity, and task performance. Our findings highlight design implications for using LLM-based AI mentors to enhance CPS learning in educational environments.

CCS Concepts

• **Human-centered computing** → **Interactive systems and tools.**

Keywords

Generative AI, Agent, creative problem solving, mentor

ACM Reference Format:

Siyu Zha, Yujia Liu, Chengbo Zheng, Jiaqi Xu, Fuze Yu, Jiangtao Gong, and Yingqing Xu. 2025. Mentigo: An Intelligent Agent for Mentoring Students in the Creative Problem Solving Process. In *CHI Conference on Human Factors in Computing Systems (CHI '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 22 pages. <https://doi.org/10.1145/3706598.3713952>

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ACM ISBN 979-8-4007-1394-1/25/04

<https://doi.org/10.1145/3706598.3713952>

Factors in Computing Systems (CHI '25), April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 22 pages. <https://doi.org/10.1145/3706598.3713952>

1 Introduction

Creative Problem Solving (CPS) is a well-established instructional method that encourages students to develop creative and critical thinking by addressing complex real-world problems collaboratively [68, 86, 110]. It has been shown to significantly enhance student engagement, creativity, and active learning [6, 49, 78, 83]. In middle school, CPS projects often involve tasks like designing sustainable solutions to local issues, creating innovative prototypes, or addressing community needs [17, 109]. However, implementing CPS in educational settings presents unique challenges, especially in middle school, where students require more tailored guidance to navigate interdisciplinary tasks and sustain engagement [18, 65, 73, 111]. Teachers must provide continuous and customized mentorship to ensure educational quality, yet doing so at scale is resource-intensive and impractical for most schools [22, 64]. This demand for increased mentorship, particularly in STEM education aimed at cultivating critical and creative thinking [56, 73], underscores the need for innovative solutions to support middle school students in their CPS learning journey.

Generative Artificial Intelligence (GAI) offers an opportunity to address these challenges by acting as a scalable, personalized mentor for middle school CPS projects. GAI applications, such as ChatGPT, have garnered attention for their ability to enhance computational, creative, and critical thinking skills [7, 36]. Unlike traditional rule-based tutoring systems, GAI can adaptively guide students through each stage of the CPS process, providing real-time feedback and scaffolding while reducing the workload on human teachers [21, 69]. Despite this potential, most studies on AI in education have focused on younger children or adult learners [70, 103], leaving the unique needs of middle school students underexplored. Furthermore, while previous research has positioned GAI primarily

as a tool for task assistance [50, 53], this study explores its role as a mentor capable of fostering deeper learning, skill development, and personalized support for middle school students. This paper contributes to the growing discussions in human-computer interaction (HCI) and educational technology by introducing an evidence-based exploration of GAI as a mentoring agent specifically designed to support middle school CPS projects, filling an important gap in the current literature.

To address these gaps, this study introduces Mentigo, an interactive AI-driven system designed to mentor middle school students throughout the CPS process. Mentigo combines structured guidance, adaptive feedback, and empathetic interactions to provide tailored support for students. The system operates through a six-stage CPS process, where it sets clear task objectives and progression standards at each stage. Mentigo dynamically adapts its mentoring strategies by monitoring students' cognitive and emotional states, utilizing a comprehensive database that matches these states with contextually relevant guidance strategies. This database-driven approach enables Mentigo to deliver personalized, flexible educational experiences that are responsive to the unique needs of each learner, fostering deeper engagement and critical thinking during the CPS tasks. We conducted a user study to evaluate Mentigo's effectiveness compared to a baseline system. The study showed that Mentigo significantly enhanced student engagement and promoted deeper cognitive processes, such as analysis and creativity, by providing structured guidance and adaptive feedback. Participants noted that the system's ability to offer personalized, reflective guidance improved their understanding and application of CPS skills.

In summary, the contributions of this study are as follows:

- Development of Mentigo, an interactive system that supports a human-AI collaborative CPS process, empowering middle school students at each stage of CPS;
- Collection of real-world datasets from middle school students engaged in CPS, providing valuable insights into student-mentor interactions;
- Implementation of a comparative experiment and an expert review study to evaluate the effectiveness of Mentigo in fostering CPS skills;
- Design implications derived from the design process and user studies, offering guidelines for supporting student CPS with generative AI in educational contexts.

Overall, this study provides a valuable exploration of the application of AI mentor agents and offers insights for future research on their use in educational settings.

2 Related Work

2.1 Students' Creative Problem-Solving

Creative Problem-Solving (CPS) is a teaching method that fosters student engagement, motivation, and the development of critical and creative thinking skills essential for learners [18, 83]. CPS involves formulating problems, generating and evaluating novel solutions, and implementing effective actions [2, 98]. Studies in education and HCI have demonstrated how cognitive and computational approaches, such as socio-technical systems and AI-driven

tools, enhance CPS. For example, Hilliges et al. explored interactive brainstorming systems to facilitate collaboration, while Ali et al. demonstrated how social robots improve engagement and efficiency in problem-solving [3, 29]. Most research emphasizes digital aids that assist users in creative tasks, following structured CPS frameworks like those proposed by Treffinger and Gizzi, which include stages such as problem understanding, idea generation, and action planning [21, 100]. In this study, we adopt a six-stage CPS process: "Problem Discovery," "Information Collection," "Problem Definition," "Solution Ideation," "Solution Evaluation," and "Solution Implementation."

Despite its potential, implementing CPS projects in classrooms presents several challenges. CPS requires students to develop engagement, innovation, and critical reasoning abilities as they collaboratively investigate and solve problems, acquiring necessary knowledge and skills along the way [83]. Moreover, teachers face the challenge of guiding students in group work, planning, problem-solving, idea testing, and peer presentations, often within constrained timeframes [64]. Various roles support CPS tasks, including tutors, mentors, and teaching assistants (TAs), each with distinct approaches. Tutors focus on helping students master specific academic content or skills and facilitate an effective learning environment within small groups [38]. TAs assist with planning, preparing, and supporting teaching activities, such as grading and classroom management [43]. In contrast, this study adopts the concept of mentoring, which provides comprehensive support through learner-centered relationships often involving counseling, friendship, and socialization [11, 26, 48]. Researchers advocate for mentoring in teaching to empower students to engage more deeply in CPS tasks, allowing both mentor and mentee to develop CPS skills together [46, 108]. Mentoring has also been widely recognized for enhancing both teacher development and student learning [32].

Providing adequate human mentorship in classroom settings can be challenging due to resource constraints. AI offers a promising alternative for filling this gap [21]. Intelligent Tutoring Systems (ITSs) have demonstrated consistent learning gains over decades [44, 55, 90]. Recent studies are exploring the role of mentor agents in learning environments [4, 69]. However, there is still a lack of research on the application of generative AI specifically in the context of CPS. Our study seeks to address this gap by designing a mentor agent to guide students through the CPS process. Through user studies, we aim to explore how an AI agent can aid students in solving real-world problems and developing their CPS skills.

2.2 Generative AI for Student Learning

The integration of generative AI into education is progressing cautiously, with structured strategies and guidelines still emerging [47, 80, 82]. While some researchers advocate for its transformative potential, such as providing personalized resources, step-by-step solutions, and enriching educational practices [1, 39, 54]. Others raise concerns about misuse, particularly in areas like online assessments where tools like ChatGPT can produce essays with advanced critical thinking, sparking debates on academic integrity [92]. Despite these differing perspectives, generative AI is increasingly recognized as a vital support in education, offering

significant opportunities to enhance learning while underscoring the need for careful oversight and ethical considerations [71, 105].

Besides, many researchers proposed that generative AI can effectively assist in curriculum exercises to empower students' CPS [5, 93]. Recent studies are highlighting LLMs, particularly GPT-4, showing that they can be powerful tools to support CPS [12, 23]. Noy et al. found that ChatGPT can enhance the quality of creative solutions [72]. Hwang, A. H. C., & Won, A. S. investigated how human subjects collaborate with a computer-mediated chatbot in creative idea-generation tasks [35]. Douglas et al.'s research points out that LLMs can do even better than traditional brainstorming methods and average human performance in specific tasks when given the right prompts and paradigms [91, 107]. Therefore, researchers recommended designing AI-involved learning tasks that engage students in real-world problem-solving to meet learning objectives effectively [106].

In conclusion, generative AI has demonstrated the potential to improve students' CPS. However, their actual use, especially involving generative AI in the CPS process, remains relatively rare. Our study aims to design an agent for mentoring students in the CPS process, with a focus on middle school students. Our goal is to offer insights into how generative AI can be effectively incorporated into educational settings, empowering students in each stage of CPS in middle school.

3 Study Procedure

Our study was driven by three key motivations: (1) Identifying CPS Support Needs: Insights from experienced educators helped uncover challenges in CPS teaching, informing the design of an effective educational agent. (2) Building an Educationally Grounded System: We developed a dataset-driven AI mentorship framework aligned with CPS principles, ensuring usability and real-time interaction. (3) Evaluating System Effectiveness: We conducted a controlled user study and gathered expert feedback to assess the system's impact on CPS skills, user experience, and learning outcomes. These motivations guided the three-phase research process outlined in Figure 1.

We followed a three-phase research process to design and evaluate Mentigo. Phase 1 involved semi-structured interviews with five experienced educators to identify CPS challenges and define five design goals. Phase 2 focused on building a comprehensive system framework by structuring mentor-student interaction data into stages, states, and strategies, informing Mentigo's database, controller agent, and mentor agent. Phase 3 evaluated Mentigo through a within-subjects study with 12 middle school students, comparing its performance against a baseline system. Additionally, five expert educators reviewed the system, providing feedback to refine its design. This iterative process ensured that Mentigo's development was grounded in educational practices and real-world insights.

4 Formative Study with Experienced Educators

To better understand the challenges faced by educators and students in CPS tasks, we conducted a formative study consisting of in-depth interviews with experienced teachers and mentors. The goal of this study was to identify key areas where support is needed,

thereby informing the design of an AI-based mentoring agent to assist in the CPS process. The interviews provided valuable insights into the need for timely feedback, diversified guidance strategies, personalized support, and sustained motivation mechanisms in educational settings.

4.1 Participants and Procedure

Five experienced educators participated in this formative study, including three secondary school teachers who are in charge of CPS projects at their schools and two university mentors with backgrounds in education and extensive experience in mentoring middle school students. The three teachers have rich experience in designing and implementing CPS in their respective schools, while the two mentors have participated in various CPS projects for middle school students and have gained insights into student guidance.

The interviews were conducted via Tencent Meeting. The process began by collecting basic background information from participants, including their teaching or mentoring experience, years of practice, and current roles. During the interviews, the teachers were asked about their experiences in different phases of CPS implementation, including the specific strategies they employed, challenges encountered, and their use of AI tools to assist in teaching. We also explored their perspectives on how an AI agent could potentially support these projects in the future. Mentors were further asked about their experiences in guiding students, focusing on the types of questions commonly posed by students, the feedback they provide, and the challenges they observe in student performance. Additionally, they offered suggestions for improving guided strategies to better support students in CPS tasks.

4.2 Findings

The findings below summarize the key areas where support and improvements are needed to enhance the teaching and learning experience in the CPS process.

Need for Structured Guidance: Teachers emphasized the importance of providing structured guidance to help students navigate complex CPS tasks with a clear sense of direction and understanding of each stage's specific objectives. They noted that students need explicit instructions that clearly outline the tasks and expectations for each phase to ensure focused progress. Teachers also pointed out that it is challenging for them to constantly check in with each student to monitor their understanding and task alignment, making a structured and well-defined guidance system crucial for effective learning.

Importance for Diverse Strategies: The study emphasized the need for a variety of guidance strategies to accommodate different learning styles and needs. They suggested incorporating multiple approaches, such as task planning, creative stimulation, and in-depth analysis, to effectively engage students and enhance their CPS skills. The interviewed mentors highlighted the effectiveness of scaffolding techniques and open-ended questions to encourage independent thinking and learning. Since CPS involves both divergent and convergent thinking stages, it is crucial for teachers to provide different types of guidance tailored to each phase to help students progressively acquire CPS-related knowledge and

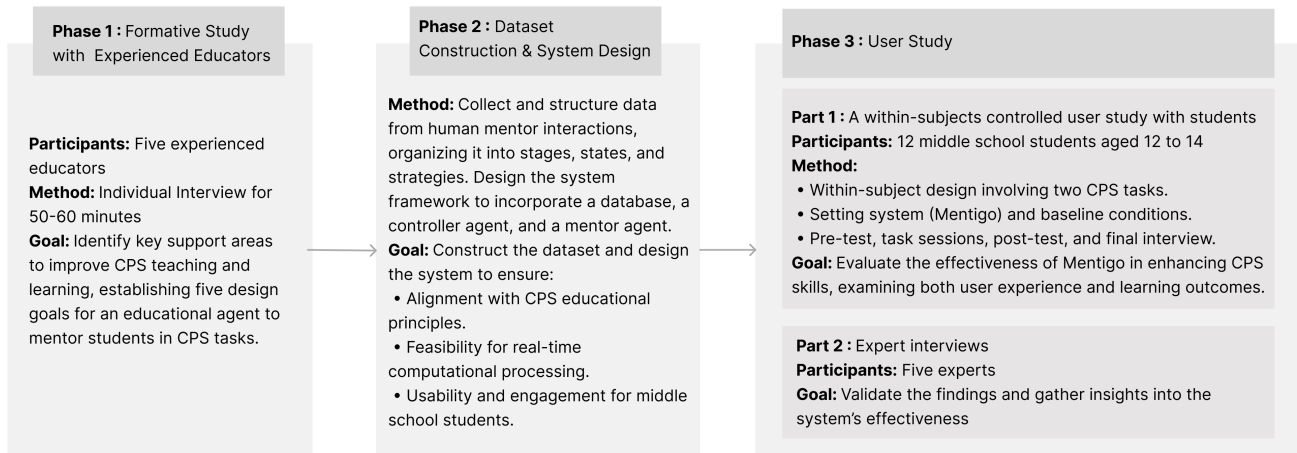


Figure 1: Three-phased study procedure

skills. Additionally, mentors pointed out the importance of specific feedback mechanisms and continuous evaluation to better support students' growth and development throughout the process.

Value of Personalized Feedback: The study highlighted the importance of providing personalized feedback based on real-time student performance, as feedback tailored to a student's specific progress and needs is more effective in promoting learning and sustaining engagement. Teachers noted that students often require timely and detailed feedback throughout each stage of their CPS tasks, relying on precise, process-oriented guidance to navigate complex challenges. However, due to limitations in time and resources, it is difficult for teachers to deliver one-on-one personalized feedback to every student, particularly on professional and procedural aspects. This challenge is even more pronounced for novice teachers, who may struggle to provide effective guidance. In this context, educators suggested introducing a supportive role, such as a mentor, to offer timely and personalized feedback at various stages of project implementation.

Encouraging Deep Cognitive Engagement: Teachers emphasized the importance of encouraging students to engage in deeper cognitive processes and critical thinking. They suggested that the educational agent should prompt students to explore problems thoroughly, consider various perspectives, and reflect on their reasoning to develop comprehensive solutions. Teachers face significant difficulties in prompting students to engage in deeper thinking, especially in helping them evaluate and justify the value and significance of their proposed solutions. Teachers widely believe that a mentor, providing one-on-one feedback, guidance, and direction, would greatly improve the teaching effectiveness in CPS tasks.

Need for Emotional Support: Teachers emphasized the necessity of providing emotional support to students throughout CPS tasks. They highlighted that empathetic interactions, encouragement, and an understanding of students' emotional states are vital for maintaining motivation and fostering a positive learning environment. Educators pointed out that students often face frustration

and anxiety during complex problem-solving activities, which can hinder their learning progress. Therefore, incorporating emotional support through a mentor who can recognize emotional cues and respond with empathy could help sustain students' engagement and confidence, ultimately contributing to a more supportive and effective educational experience.

These findings outline the various needs of students in CPS tasks and emphasize the importance of mentors in providing effective support to address these needs in the design of agents.

4.3 Design Goals

Based on the findings from our formative study (as detailed in subsection 4.2), we establish five design goals (DGs) essential for developing an educational agent that mentors students in CPS tasks.

DG1: Ensure Structured Task Management for Clarity and Compliance. In CPS, middle school students need structured task management and phased guidance to understand the specific requirements and expected outcomes at each stage [59]. Our findings emphasize the need for clear and explicit task guidance to help students stay focused and aligned with learning objectives. Therefore, the educational agent must incorporate rigorous task management features that clearly define and reinforce the objectives and requirements for each CPS phase [13]. This structured approach will help students adhere to their learning goals and improve overall CPS performance by reducing ambiguity and enhancing task clarity.

DG2: Diversify Guided Strategies for Enhanced CPS Skills. Effective CPS requires students to engage in deep, creative, and critical thinking, which is best supported through diverse guided strategies [51, 99]. Our findings suggest that employing a variety of pedagogical approaches, such as scaffolding techniques, open-ended questioning, and problem-based learning, can help students develop robust knowledge frameworks and tackle complex problems. Therefore, the educational agent should be designed to utilize a wide range of guided strategies that are adaptable and tailored

to different problem-solving contexts, ensuring that guidance is dynamic and responsive to students' evolving needs.

DG3: Provide Personalized Feedback through Dynamic Student State. Students require timely and personalized feedback throughout their CPS tasks, where they heavily depend on specific, process-oriented guidance [10, 96]. However, teachers often struggle to provide one-on-one personalized feedback due to time and resource constraints. To address this need, the educational agent should develop mechanisms that continuously assess and interpret the student's states, enabling the delivery of contextually appropriate and individualized feedback. This approach would enhance the student learning experience and support their continuous development.

DG4: Encourage Deep Cognitive Engagement and Conceptual Mastery. Maintaining student motivation during challenging creative tasks is essential for fostering deep cognitive engagement and achieving conceptual mastery [83, 85]. Research indicates that students benefit from consistent encouragement to engage in critical thinking and reflective practices. Therefore, the educational agent should be designed to actively promote deep cognitive engagement by encouraging continuous reflection and critical analysis, guiding students to internalize and effectively apply CPS methodologies. This would help sustain interest, foster deeper understanding, and enhance long-term retention of concepts.

DG5: Integrate Empathetic Interactions to Foster Emotional Engagement. Maintaining student motivation and emotional well-being is crucial, particularly during prolonged creative tasks where frustration and disengagement can occur [62]. Studies show that students benefit from empathetic support that addresses both their emotional and academic needs. To create a supportive learning environment, the educational agent should incorporate emotionally intelligent responses that simulate human-like empathy, helping to nurture both the student's emotional resilience and academic success. These design goals, grounded in research findings, provide a comprehensive framework for developing an educational agent that effectively supports students in CPS tasks through dynamic, adaptive, and empathetic learning interventions.

5 The Mentigo Agent System

5.1 System Overview

In this section, we present the design of **Mentigo**, an LLM-based agent framework developed to act as a personalized mentor for students working on CPS tasks. The system framework (see Fig. 2) comprises three main components: a database, a controller agent, and a mentor agent, integrated with a user interface to support interaction. These components were designed to achieve the system's five design goals (DG1-DG5). To address DG1-DG3, we developed a comprehensive database that encodes human mentor data into three parts: Stage, State, and Strategy. Each entry includes names, descriptions, examples, and mappings between states and strategies. For instance, the state "misunderstanding of task" maps to strategies such as "restate task" and "provide examples." This structured organization allows the system to retrieve relevant strategies dynamically, providing contextually accurate guidance aligned with educational practices.

The controller agent supports DG4 by handling logical decision-making. When a student inputs a query, the controller agent analyzes the dialogue history and performs three tasks: determining the current CPS stage, updating the student's state, and identifying the most urgent state along with the corresponding best strategy. These structured parameters guide the agent's subsequent actions. The mentor agent, designed for DG5, uses these parameters to generate empathetic, context-specific guidance. Prompts include the mentor's persona, stage-specific objectives, and detailed descriptions of the problem and strategy. This ensures the guidance is not only informative but also engaging and supportive. The system is deployed on a web-based platform, enabling real-time text or voice interactions between students and the mentor agent. The interface tracks dialogue history, allowing students and mentors to review progress. By combining these components, **Mentigo** dynamically adapts to student needs, providing personalized and structured mentorship throughout CPS tasks while fostering engagement and critical thinking.

5.2 Mentigo Dataset Construction

This study focuses on extracting student states and mentor guidance strategies from authentic CPS scenarios to inform the design of our agent system. We conducted two CPS workshops involving 57 middle school students, divided into six groups per workshop, each guided by one of 12 mentors with backgrounds in education, technological development, or engineering. These workshops replicated classroom dynamics to capture interactions across the six stages of CPS. A total of 80 hours of classroom sessions were recorded, transcribed, and coded, resulting in 1,615 dialogue entries and 747 instances of guidance strategies and student responses. Drawing from research on the cognitive and emotional dynamics of learning [52, 75, 87] and mentoring strategies in education [15, 33, 63], we developed a categorization framework to classify student states (e.g., confusion, confidence) and mentor strategies (e.g., probing, encouragement). Studies highlight key states such as confusion, which signals cognitive engagement, and confidence, which correlates with persistence and success [75]. Research on mentoring strategies emphasizes scaffolding techniques and encouragement to foster self-regulation and transfer of knowledge [33, 63]. This iterative coding process, informed by both theoretical frameworks and emergent patterns in our data, produced a comprehensive dataset that captures the subtleties of real classroom interactions. This dataset serves as a foundational element for developing our agent system, enabling it to understand various student states and apply appropriate mentoring strategies. Based on this dataset, we achieve the first three design goals (DG1-DG3) as follows:

5.2.1 [DG1]Ensure Structured Task Management for Clarity and Compliance. To mentor students through CPS tasks clearly and effectively, the Mentigo system employs a structured task management approach using dynamic learning phases. The system divides the learning process into six distinct phases: "Problem Discovery," "Information Collection," "Problem Definition," "Solution Ideation," "Solution Evaluation," and "Solution Implementation." Each stage has clearly defined tasks, descriptions, and expected outcomes managed through a predefined "Stage".

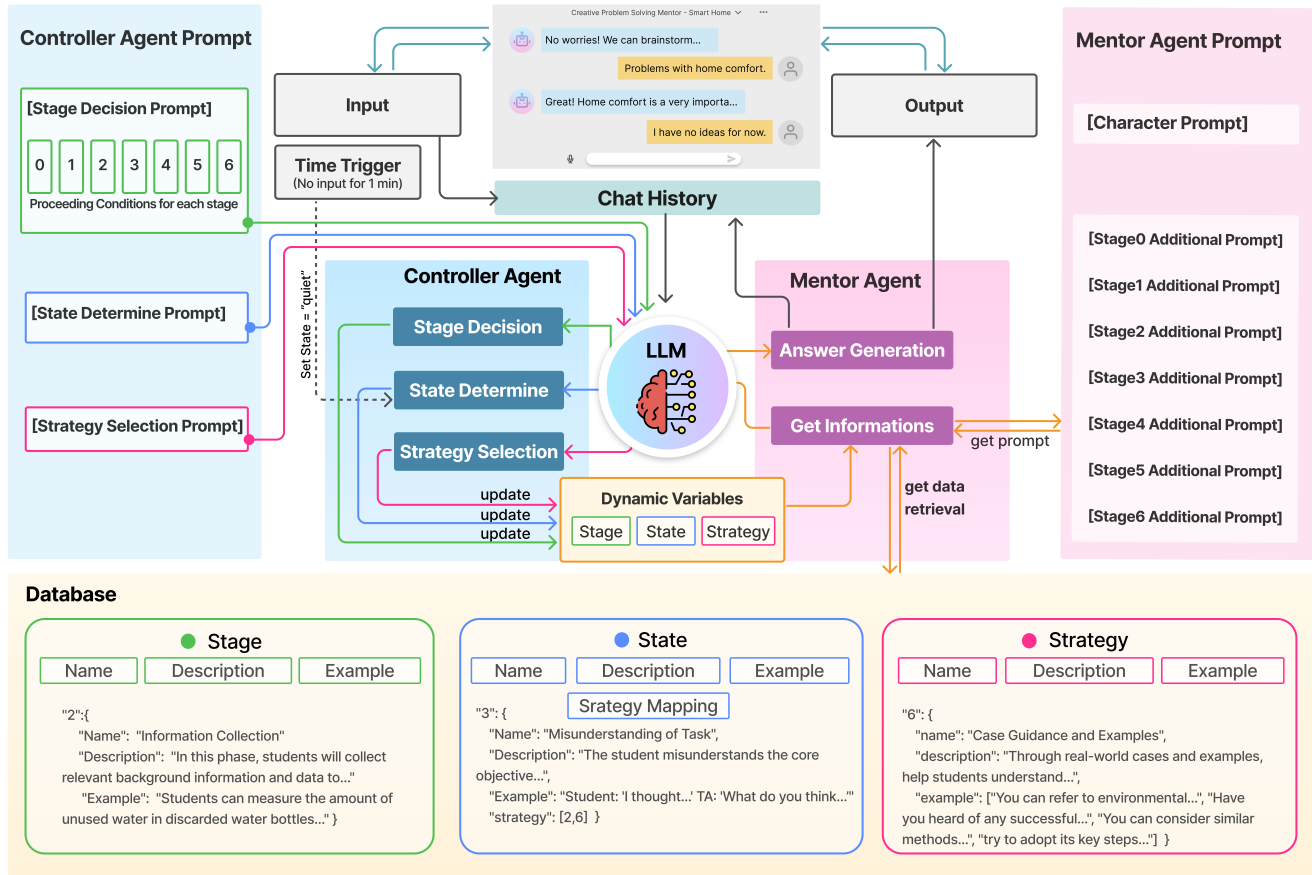


Figure 2: Framework of Mentigo Agent. The Controller Agent manages the flow of student interactions through three functions: Stage Decision, State Determine, and Strategy Selection. The Mentor Agent provides tailored feedback and guidance prompts based on students' progress. The Database stores information on different Stages, States, and Strategies to support adaptive decision-making, with all interactions facilitated by an LLM. The figure also includes a UI dialogue interface that visualizes the interaction process, displaying the student's input and the Mentor Agent's responses in a conversational format, alongside real-time updates on stages, states, and strategies. A UI dialogue interface presents student inputs, Mentor Agent responses, and real-time updates on stages, states, and strategies.

This structure is part of the underlying dataset that supports the controller agent's logical decision-making. Table 1 provides a comprehensive overview of each phase and its definitions, which serves as the foundation for the system to manage task progression and ensure that students are guided step-by-step. Grounded in educational theories that emphasize phased learning for complex tasks, this structured approach helps minimize cognitive overload and ensures students gain a thorough understanding of each phase's objectives. Maintaining focus on learning goals enhances task compliance and overall student performance.

5.2.2 [DG2] Diversify Guided Strategies for Enhanced CPS Skills.

The system incorporates a range of guided strategies to

provide adaptive learning support and enhance CPS skills. This approach is informed by pedagogical frameworks such as scaffolding and cognitive tools, which are known to promote deeper engagement and critical thinking in learners. Using a predefined "Stage," the system dynamically selects appropriate guided strategies based on students' cognitive and emotional states, ensuring that the guidance provided is contextually relevant and effective.

To systematically develop a robust set of guided strategies for CPS tasks, we employed a structured coding process based on authentic classroom interactions. We began by identifying a broad range of potential strategies from recorded dialogues between mentors and students during various CPS stages. Three researchers

Table 1: Stages of Creative Problem Solving (CPS)

CPS Stages	Definition
Problem Discovery	In this stage, students identify potential problems or challenges related to the project theme. Through observing the campus environment, discussions, or guidance, students can find practical issues worth attention. The goal of this stage is to help students discover opportunities for improvement related to the project theme around them.
Information Collection	In this stage, students gather relevant background information and data to better understand the problems they have discovered. They may conduct surveys, collect data, or talk to relevant individuals to obtain information related to the project theme.
Problem Definition	Based on the data and information collected, students define the core problem. This stage helps students clarify what needs to be solved and ensures they focus on the truly significant challenges.
Solution Ideation	Students generate multiple potential solutions through brainstorming and creative thinking. They are encouraged to propose a variety of innovative ideas, regardless of their initial practicality.
Solution Evaluation	In this stage, students evaluate the solutions they have generated and select the most effective and feasible one. They will consider factors such as feasibility, cost, potential impact, and how to gain more support and participation for the solution.
Solution Implementation	In this stage, students complete a final implementation plan report, which should include an introduction to the background of the chosen solution, the conception of the solution, a specific implementation plan, and potential challenges that may arise during the implementation process.

independently coded these dialogues to extract specific mentoring strategies, resulting in 20 distinct guided strategies that reflect the diverse techniques used by mentors to support student learning. Each strategy was then clearly defined to capture its unique characteristics and application.

These 20 strategies were subsequently organized into five overarching categories: "Task Planning and Execution," "Creative Thinking Stimulation," "In-depth Thinking and Systematic Analysis," "Information Integration and Resource Support," and "Emotional Support and Motivation." This categorization provides a comprehensive framework for understanding how different strategies can be applied effectively in CPS contexts. Each strategy was also aligned with specific student states identified earlier, enabling targeted interventions. The strategy vocabulary was further refined through feedback from experienced mentors and educational experts to ensure it is practically relevant. The final set of 20 strategies, grouped into five major categories, is detailed in Table 2 and serves as the foundation for developing our system, allowing it to simulate effective mentoring by dynamically applying strategies based on student states and learning needs.

5.2.3 [DG3]Provide Personalize Feedback through Dynamic Student State. Mentigo continuously monitors students' progress to deliver personalized feedback by interpreting real-time inputs related to engagement, performance, and emotional states. Drawing from our real-world dataset analysis, we identified common student states in CPS tasks and matched them with appropriate guidance strategies. For instance, if a student is "stuck," the system provides "case-based guidance," whereas for a student "lacking confidence," it offers encouraging feedback.

We developed a comprehensive set of student state codes to analyze learning states in CPS scenarios. The process began with an initial screening of classroom recordings to identify relevant student states associated with CPS activities. Three researchers independently coded the dialogues, resulting in 23 specific student states that represent a range of behaviors and emotional responses during CPS tasks. These states were refined into eight broader

categories, such as "Time Management Challenges" and "Emotional Support Needs," providing a clearer framework for understanding student engagement.

Each student state is linked to corresponding guidance strategies to address specific needs, making the dataset both descriptive and actionable. For example, the state "Confused Goals" is associated with strategies like task clarification. The vocabulary was further refined with input from mentors and educational experts to ensure it captures the full spectrum of student behaviors. The final dataset, comprising 23 student states categorized into eight major categories with definitions and associated strategies (see Table 3 for details), serves as a foundation for the agent system to identify student states and apply the appropriate guidance strategies during CPS activities.

5.3 Mentor Agent Workflow

5.3.1 [DG4]Encourage Deep Cognitive Engagement and Conceptual Mastery. To better manage the CPS process and counteract the potential instability of large language models, we developed a controller agent that uses structured outputs to strictly manage three dynamic parameters: stage, state, and strategy. The controller agent operates solely in a decision-making capacity without direct interaction with users, ensuring consistency and clarity in guiding students through each phase of the CPS process.

The system begins by prompting the large model for a "stage decision" after each dialogue round. Each stage has strict criteria that must be met (see Fig. 3, Stage Decision Prompt), and only when all requirements are fulfilled can the process advance to the next stage (stage +1). Otherwise, the process remains in the current stage, and the "stage" parameter is updated accordingly. This method ensures that students fully comprehend each phase's requirements before moving forward.

Simultaneously, the system determines the "student state" by analyzing real-time inputs. If the stage advances, the state is updated to "stage start." If not, the system provides the model with possible student states, their definitions, and the chat history to assess the

Table 2: Taxonomy of teachers' strategies for guiding students in CPS

ID	Guidance Category	Specific Strategy
1	Task Planning and Execution Support	Task Follow-up and Allocation
2		Task Restatement
3	Creative Thinking Stimulation	Brainstorming Guidance
4		Random Association and Creative Stimulation
5		Creative Filtering and Focusing
6		Case Guidance and Examples
7		Challenge Guidance
8	Deep Thinking and Systematic Analysis	Probing Guidance
9		System Thinking Training
10		Critical Thinking Practice
11		Deep Thinking and Reflection
12		Rebuttal and Debate
13	Information Integration and Resource Support	Resource Guidance
14		Content Review
15		Information Collection and Classification
16		Information Integration Guidance
17		Scenario Simulation and Feedback
18	Emotional Support and Motivation Stimulation	Encouragement and Motivation
19		Emotional Counseling
20		Immediate Feedback

student's current states, excluding "stage start" and "quiet" states. The "quiet" state is a special condition triggered by inactivity (e.g., no user input for one minute) to encourage proactive guidance. The updated "state" parameter list is then used to identify which strategies are most suitable for the student's current situation.

Once the stage and state parameters are updated based on the current dialogue, the system maps each state to corresponding guidance strategies. The model autonomously determines which student state requires immediate attention and selects the most appropriate strategy for that state. This process results in a dynamically chosen "strategy" that promotes deep cognitive engagement and supports students in mastering CPS concepts. This adaptive approach, grounded in real-time assessment and structured prompts, ensures that the agent remains responsive to individual student needs and provides targeted, context-aware guidance.

5.3.2 [DG5] Integrate Empathetic Interactions to Foster Emotional Engagement. To enhance emotional engagement and create a more supportive learning environment, we designed a mentor agent specifically to guide students empathetically through their CPS tasks, as shown in Fig. 4. The mentor agent is structured around two key components: the "character prompt" and the "additional prompt for each step." The "character prompt" defines the mentor agent's overall persona, tone, and communication style, providing a consistent framework for interaction that is input into the system at every dialogue round. This persona is designed to be approachable, supportive, and motivational, creating a positive emotional connection with the student.

The "additional prompt for each step" serves as a supplementary guide tailored to each phase of the CPS process, offering more specific instructions and examples that align with the current stage's objectives. These prompts help the mentor agent provide relevant

and context-aware guidance that encourages students to stay engaged, reflect deeply, and think critically about their tasks. By adjusting the prompts according to the requirements of each phase, the system ensures that the mentoring is both effective and aligned with pedagogical goals.

Moreover, we aim for Mentigo to dynamically adjust its guidance strategies based on the student's evolving states, as determined by the Controller Agent. Using the parameters provided by the Controller Agent—such as the current stage, student state, and chosen strategy—the system accesses the database to retrieve appropriate strategies and example dialogues. These examples are crafted to make the mentor agent's responses more precise, concise, and natural, mirroring how a human mentor would communicate. All relevant information is then input into the mentor agent, which synthesizes this data to generate the final output, ensuring that each interaction is empathetic and tailored to the student's immediate needs. This design allows the mentor agent to not only guide students effectively but also to build an emotional rapport that fosters deeper engagement and motivation throughout the learning process.

5.4 System Implementation

5.4.1 Dataset Construction. From the 80 hours of classroom recordings involving 57 middle school students and 12 mentors across two workshops, we constructed a comprehensive dataset of 2,362 interaction sequences between students and mentors. These interactions were carefully coded into 23 specific student states, grouped under eight categories, and 20 mentor strategies, categorized under five guidance categories.

The dataset captures the dynamic nature of real-world classroom interactions, reflecting the diverse states of students and the adaptive strategies used by mentors. For instance, when a student shows confusion about a task's goals, a mentor might respond by restating the task and providing role guidance. Such examples demonstrate

Table 3: Students' states in CPS and potential teacher guidance strategies

ID	Student Status	Specific Status	Corresponding Guidance Strategies
1	Task/ Goal Definition Unclear	Silent	Task Assignment, Restate Task
2		Goal Unclear	Task Assignment, Restate Task, Probing Guidance
3		Task Misunderstanding	Restate Task, Case Guidance and Examples
4		Goal Prioritization Confusion	Task Assignment, Restate Task
5	Time Management and Task Breakdown Difficulties	Poor Time Management	Task Assignment, Resource Guidance
6		No Ideas/Task Breakdown Difficulties	Brainstorming Guidance, Encouragement and Motivation, Immediate Feedback
7	Lack of Creative and Divergent Thinking	Limited Thinking	Creative Flow Selection and Focus, Random Association and Creative Stimulation, Scenario Simulation and Scene Feedback, Case Guidance and Examples
8		Insufficient Ideas	Random Association and Creative Stimulation, Encouragement and Motivation, Deep Thinking and Reflection
9		Overly Divergent Thinking	Creative Flow Selection and Focus, Refutation and Debate, Systematic Thinking Training
10	Insufficient In-depth Thinking and Systematic Analysis	Shallow Thinking	Challenge Guidance, Systematic Thinking Training, Probing Guidance
11		Lack of Critical Thinking	Critical Thinking Exercises, Refutation and Debate, Scenario Simulation and Scene Feedback
12		Lack of Systematic Thinking	Systematic Thinking Training, Deep Thinking and Reflection, Critical Thinking Exercises
13	Insufficient Information Collection and Integration Skills	Inadequate Information Collection	Information Gathering and Classification, Resource Guidance, Content Review
14		Inadequate Information Analysis Skills	Information Integration Guidance, Refutation and Debate, Case Guidance and Examples
15		Inadequate Information Integration Skills	Information Integration Guidance, Systematic Thinking Training, Probing Guidance
16	Evaluation and Decision-Making Difficulties	Lack of Evaluation Criteria	Case Guidance and Examples, Critical Thinking Exercises, Deep Thinking and Reflection
17		Decision Uncertainty	Resource Guidance, Task Assignment, Systematic Thinking Training
18	Lack of Motivation and Confidence	Insufficient Intrinsic Motivation	Encouragement and Motivation, Immediate Feedback
19		Lack of Confidence	Encouragement and Motivation, Emotional Support, Case Guidance and Examples
20	Fatigue/ Burnout/ Negative Emotions	Learning Fatigue	Encouragement and Motivation, Emotional Support
21		Low Mood	Encouragement and Motivation, Emotional Support
22		Conflicts Arising	Intergroup Encouragement, Role Guidance, Immediate Feedback
23		Normal Progress/ Active Questioning	Encouragement and Motivation, Immediate Feedback

the variability and contextual nuances in student-mentor dialogues, providing valuable data for simulating effective CPS guidance.

To train a robust classification model for Mentigo, we expanded our dataset by involving educational experts to annotate dialogue segments, including student queries, responses, and mentor guidance. Each of the 23 student states and 20 mentor strategies was validated across multiple scenarios to ensure consistency and accuracy. This comprehensive annotation process enriched the dataset, enabling Mentigo to provide context-aware, tailored guidance in real-time classroom interactions.

In summary, the constructed dataset of 2,362 interaction sequences, with detailed annotations for student states and mentor strategies, forms the foundation for developing a data-driven agent.

This dataset equips the agent to dynamically recognize and respond to different learning states and instructional needs in CPS environments.

5.4.2 Technical Implementation Details. Our system is designed to provide dynamic, context-aware mentorship through an integration of robust frameworks and advanced AI capabilities. The implementation combines frontend, backend, and data processing technologies to ensure seamless functionality.

We utilized the Streamlit framework for rapid development of interactive web applications, enabling an integrated workflow that combines frontend and backend development directly from Python scripts. For data management, we adopted Google Sheets as a serverless backend to store user information, conversation logs, and other

Controller Agent Prompts

[Stage Decision Prompt]

"Stage 0"	"Stage 1"	"Stage 2"	"Stage 3"	"Stage 4"	"Stage 5"	"Stage 6"
Once you've collected the student's information... ...	In the "Problem Discovery" phase, the student cannot proceed if: (1) Fewer than 3-5 relevant problems are submitted. (2) Problem statements lack scenario, subject, or	In the "Information Collection" phase, the student proceed if: (1) Completed relevant information search.... (2) Written a status report on the...	In the "Problem Definition" phase, the student proceed if: (1) Submitted one problem statement ... (2) The statement includes scenario ...	In the "Solution Ideation" phase, the student proceed if: (1) Proposed three feasible ... (2) Each solution includes ...	In the "Solution Evaluation" phase, the student proceed if: (1) Evaluated the three solutions ... (2) The report must include at least ...	In the "Solution Implementation" phase, the student proceed if: (1) Complete implementation plan ... (2) The report includes background on the ...

Currently in phase: Stage.Name The current phase is defined as Stage.Description, examples are Stage.Example
 The next phase is (Stage+1).Name, defined as (Stage+1).Description, examples are (Stage+1).Example
 Based on the dialogue information, determine whether to proceed to the next phase. If staying in the current phase, set the output parameter Stage to (stage); otherwise, set the output parameter Stage to (stage+1).

[State Determine Prompt]

If not moving to the next phase, the student's status may be: State.Name State.Description
 The student's previous status list is State. Remove all '0' and '1', and based on the new dialogue record, update and output the student's status list state.
 If moving to the next phase, the student's status will be "start," and update and output the student's status list as state=[0].

[Strategy Selection Prompt]

The current student statuses and corresponding strategies are as follows: State.Name State.StrategyMapping Strategy.name Strategy.Description
 Based on the dialogue information, output the most urgent problem ID as urge_state and the corresponding best strategy ID as strategy

Controller Agent Pipeline

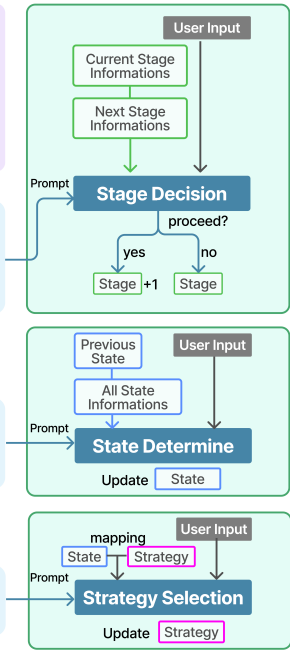


Figure 3: Prompts for Controller Agent

Mentor Agent Prompts

[Character Prompt] [Additional Prompt for each stage]

	[Stage 0]	[Stage 1]	[Stage 2]	[Stage 3]	[Stage 4]	[Stage 5]	[Stage 6]
Character Prompt	You are Mentigo, a professional, patient, and creative mentor. Your role involves actively guiding students through problem-solving.....						
Additional Prompt for each stage	Start by introducing yourself and the project theme. Ask if the student has any background knowledge or project experience.....	In the "Problem Discovery" phase, your task is to guide the student to identify 3-5 real issues related to smart home systems. Each issue must be clearly stated in one ...	In the "Information Collection" phase, your task is introduce the task goal: students need to gather relevant information based on their proposed questions and write a.....	In the "Problem Definition" phase, your task is introduce the task goal: students need to identify the final problem they want to solve based on the questions and	In the "Solution Ideation" phase, you need to: (1) Introduce the task goal: students must propose three original solutions.....	In the "Solution Evaluation" phase, you need to: (1) Introduce the task goal: students must evaluate the three solutions.....	In the "Solution Implementation" phase, you need to: (1) Introduce the task goal: guide students in completing the final report.....

[Providing Informations Prompt]

Please resolve the student's state problem: State.Name State.Description. Use the strategy: Strategy.Name Strategy.Description. The language style example is as follows: Strategy.Example

Figure 4: Prompts for Mentor Agent

critical data in real time. Google Sheets is synchronized with the mentor agent via API, ensuring smooth data retrieval and updates. User interactions and system responses are logged and processed dynamically, enabling continuous refinement of recommendations. This approach offers both scalability and flexibility, simplifying system updates and maintenance.

The conversational AI capabilities are powered by OpenAI's GPT-4o API (version 2024-08-06), integrated with Langchain to enhance the structure and flow of interactions. To generate personalized guidance, the system leverages a dataset of 2,362 interaction sequences, which were encoded into stage-state-strategy triples and

stored as case examples in the database. Using Retrieval-Augmented Generation (RAG) technology, these examples are dynamically retrieved during real-time interactions. The controller agent processes user input and determines contextual parameters (e.g., Stage: Ideation; State: Urge; Strategy: Encouragement). It then queries the database for relevant case examples, integrates them with the user's context, and inputs this structured information into the LLM. This process ensures that the guidance remains both tailored and contextually relevant.

The system is deployed on a cloud server, ensuring accessibility via web browsers. Figures 2, 3, and 4 illustrate key aspects of

the system, including the interface, operational workflows, and main prompts. Together, these components create a cohesive framework for delivering personalized, adaptive mentorship to students engaged in CPS tasks.

5.5 Technical Evaluation

To evaluate the accuracy and reliability of Mentigo’s guidance, particularly its ability to manage potential hallucinations, we conducted a systematic test simulating interactions with different distinct middle school student personality types. These types were identified through prior literature [9, 61, 84, 101] and validated through discussions with five experienced educators involved in our formative study. Based on real classroom data collected during our study and expert input from teachers, we simulated each student type generating 2-3 representative queries at each stage of the CPS process. This approach ensured a diverse range of authentic interaction scenarios, allowing us to comprehensively assess the system’s response accuracy and adaptability. A detailed list of student personality types and question types is provided in the supplementary material.

To evaluate Mentigo’s performance, we developed a codebook based on three hallucination dimensions derived from prior research [31]: Instruction Inconsistency, Context Inconsistency, and Logical Inconsistency. Instruction Inconsistency occurs when the system provides direct answers rather than facilitating reflective thinking, deviating from its intended role as a mentor. Context Inconsistency arises when the system fails to reference previous interactions, leading to irrelevant or disconnected responses. Logical Inconsistency occurs when the system violates the expected task flow, such as skipping predefined stages in the CPS process. Two independent researchers coded each dialogue turn using a binary scheme (1 for hallucination, 0 for no hallucination), ensuring systematic evaluation. Inter-rater reliability was calculated using Cohen’s Kappa, yielding a score of 0.86, indicating a high level of agreement. After resolving discrepancies through consensus discussions, Mentigo was evaluated across six CPS task stages with 266 dialogue turns, achieving an accuracy rate of 93.75%, reflecting its ability to provide context-aware, personalized guidance while minimizing errors.

6 User Study

We conducted a controlled user study under a within-subject design to evaluate the effectiveness and usability of Mentigo in supporting middle school students during CPS tasks. This study aimed to understand how Mentigo’s contextual intelligent guidance affects user engagement, depth of reflection, and integration of feedback in creative problem-solving processes. We hypothesize that the real-time personalized guidance provided by Mentigo will be perceived as a valuable tool, enhancing students’ reflection and ideation during CPS tasks.

6.1 Baseline System Setup and Performance

To ensure that Mentigo’s performance could be objectively evaluated, we established a baseline system for comparison. The baseline design was informed by prior research on the application of GAI in education, where tools like ChatGPT are commonly used either

directly [24, 66, 113] or with simple prompt engineering [14, 28]. To ensure consistency, the baseline condition used the same interface as Mentigo, with the mentor role simulated using straightforward prompts (detailed in the supplementary material). This controlled design allowed us to isolate the effects of Mentigo’s advanced mentoring features, enabling a more rigorous evaluation of its capabilities.

To ensure a fair evaluation, we conducted the same technical evaluation described in Section 5.5 for the baseline system, simulating interactions with eight distinct student personality types across six CPS stages. During this evaluation, the baseline system generated a total of 149 dialogue turns and achieved an accuracy rate of 71.58%. Although the baseline system exhibits higher rates of hallucination compared to the Mentigo system (accuracy rate is 93.75%), we still included it in our user study for two reasons: first, it represents the state-of-the-art (SOTA) solution reported in previous literature; second, it has received numerous positive feedback in past user evaluations [14, 28]. This comparison allows us to investigate whether our carefully designed Mentigo system provides additional benefits compared to the SOTA baseline system.

6.2 Participants

We recruited 12 middle school students aged 12 to 14 (6 males and 6 females) for this study (shown in Table 4). Participants had diverse backgrounds coded as follows: CPS experience (1 = has not participated, 2 = has participated in several CST activities), AI experience (1 = no AI use, 2 = occasional use of AI), and computer usage (1 = low proficiency, 2 = moderate proficiency, 3 = high proficiency). They were recruited through social media outreach, and all participants’ parents provided informed consent. We employed a randomized within-subject experimental design, allowing each participant to interact with both a control group (using a baseline system) and an experimental group (using the Mentigo system). This design enabled a direct comparison of user experiences with and without Mentigo’s intelligent guidance.

We implemented several ethical measures to ensure the safety and comfort of child participants. Informed consent was obtained from both the children and their parents, with clear explanations of the study’s purpose and procedures. Before each session, we reconfirmed the children’s willingness to participate and reminded them of their right to withdraw at any time. A researcher was present during all sessions to monitor well-being and provide support if needed. We also verified with parents that their children had no relevant mental health concerns and clearly communicated that the AI agent was not a real person. The study was approved by the university’s Institutional Review Board (IRB).

6.3 Study Procedure and Protocol

We adopted a within-subject experimental design for this user study. Participants were asked to complete CPS tasks under two conditions in a counterbalanced and randomized order: the system condition (using our fully designed educational agent with advanced features) and the baseline condition (using a basic version of the agent with simple prompts and a consistent interface). Two different CPS tasks were prepared for participants:

Table 4: Participant's Information

ID	Gender	Age	Grade	CPS Experience	AI Experience	Computer Usage
P1	Boy	13	7	1	1	3
P2	Boy	12	7	1	2	3
P3	Boy	14	7	2	1	2
P4	Girl	12	7	2	1	2
P5	Girl	13	8	2	2	3
P6	Girl	13	8	1	1	1
P7	Girl	12	7	1	1	1
P8	Boy	14	9	2	2	2
P9	Boy	14	8	1	2	3
P10	Girl	13	8	2	2	2
P11	Boy	12	7	2	2	1
P12	Girl	12	7	2	2	2

- **Smart Home Task:** This task focuses on developing innovative solutions for smart home systems. Participants were asked to identify problems in current smart home technologies and propose creative solutions to improve user experience, safety, and sustainability.
- **Low-Carbon Campus Task:** This task aimed to encourage participants to design initiatives or strategies to promote low-carbon practices in a school campus setting, considering factors such as energy efficiency, waste management, and engagement of students and staff.

For each task, participants followed a structured process that included gathering information, brainstorming solutions, and completing a learning report outlining their proposed solutions and the rationale behind them. The study sessions were conducted remotely via Tencent Meeting. Participants were first introduced to the background and purpose of the study, completed a demographic survey, and read the consent form. After obtaining their consent, participants were guided through an introduction to the basic functionalities of the two agents and the study process. The study consisted of several phases:

- (1) **Introduction and Familiarization with the System (10 minutes):** Participants were introduced to the study's objectives and guided through the educational agent's basic features, including interface navigation, task templates, and system guidance. For those familiar with similar tools, a quick refresher ensured all participants were comfortable using the system.
- (2) **Pre-Test Questionnaire (5 minutes):** After familiarization, participants completed a pre-test questionnaire designed to measure their prior knowledge and skills related to CPS. This served as a baseline for evaluating changes in their CPS skills throughout the study.
- (3) **Task Sessions: System Condition and Baseline Condition (30-40 minutes each):** Participants were randomly assigned to complete the two tasks-Smart Home Task and Low-Carbon Campus Task-under two different conditions: the system condition using Mentigo and the baseline condition. The order of the conditions was counterbalanced to avoid learning effects.

System Condition Task (Mentigo): In this condition, participants used Mentigo, the fully designed educational agent

with advanced features such as dynamic prompts, reflective questions, and adaptive feedback. They had 30 minutes to complete the task with Mentigo's guidance, followed by completing a structured learning report.

Baseline Condition Task: Participants used a simplified system version with basic prompts and a consistent interface, lacking the advanced features of Mentigo. After a brief introduction, they had 30 minutes to complete the second task, following the same procedure as in the system condition.

- (4) **Post-Test Questionnaire for Each Condition (5 minutes each):** After completing each task, participants filled out a post-test questionnaire to evaluate the system's usability, perceived cognitive load, and mastery of CPS skills. This feedback was crucial for comparing the effectiveness of Mentigo with the basic agent setup and understanding how the advanced features influenced user experience and learning outcomes.
- (5) **Final Interview (20 minutes):** After completing both conditions, participants took part in a final in-depth interview to provide qualitative insights on their experiences with both the Mentigo and baseline systems. The interview focused on their thought processes, challenges encountered, and preferences between the two systems. This feedback was vital for understanding user interactions and identifying key areas for enhancing the system's design and functionality.

This combined procedure and protocol provided a comprehensive framework for evaluating the educational agent's effectiveness and usability across different CPS contexts.

6.4 Data Collection and Analysis

Our data came from four primary sources: 1) pre-test and post-test questionnaires to assess knowledge gain, system usability, and task load, 2) learning reports created by participants during tasks to evaluate task performance outcomes, 3) screen recordings and system logs to capture student-system interaction data, and 4) in-depth interviews to gather qualitative feedback on user experiences. The pre- and post-test data were used to compare learning outcomes and cognitive load between the Mentigo and baseline conditions. In contrast, the learning reports, interaction data, and interview data were used to analyze task performance, user engagement, and areas for system improvement.

Table 5: Manual Coding Scheme of System Data

Categories	Types	Description
Speech Type	Positive	Proactively suggest solutions to problems; proactively ask questions.
	Neutral	Express confusion or guide requests to AI
	Negative	Request AI to provide direct answers; simply copy the AI's responses.
Cognitive Level	Remembering	Reciting and memorizing labels
	Understanding	Relating and organizing previous knowledge
	Applying	Applying information into a new situation
	Analyzing	Drawing connections among ideas
	Evaluation	Examining information and make judgment
	Creation	Creating new ideas using what has just been learned

6.4.1 Pre- and Post-Test Data. To measure participants' knowledge gain, system usability, and cognitive load, pre-test and post-test questionnaires were administered before and after each task session. The pre-test assessed participants' baseline familiarity with CPS concepts, while the post-test evaluated system usability (specifically for the Mentigo condition), cognitive load, and CPS skills mastery. Knowledge gain was determined by comparing pre-and post-test results, reflecting changes in participants' self-rated knowledge on CPS tasks.

Descriptive statistics were used to summarize participants' demographics and baseline characteristics for data analysis. Paired-sample t-tests were conducted to compare cognitive load (measured using the NASA Task Load Index (TLX) [27]) and system usability between the Mentigo and baseline conditions. Repeated measures were used to identify significant differences in knowledge gain and task load, providing insights into the educational agent's effectiveness and cognitive demands associated with each condition.

6.4.2 Task Performance Data. Participants' task performance was evaluated using learning reports created during each condition (Mentigo and baseline). These reports provided insights into participants' ability to apply CPS techniques, the quality of their solutions, and their reasoning processes. To assess these reports, we developed an evaluation matrix for AI-involved CPS tasks based on Urban et al. [104]. The matrix included four dimensions: 1) Quality, 2) Elaboration, 3) Originality, and 4) Human-like Expression (see supplementary materials for details). Three researchers independently rated the reports according to these criteria, ensuring reliability through inter-rater agreement. The scores were analyzed using descriptive statistics to summarize performance in each dimension and non-parametric tests, such as the Mann-Whitney test, to compare differences between the Mentigo and baseline conditions. This provided insights into how effectively the educational agent supported CPS skills development.

6.4.3 Student-System Interaction Data. Screen recordings and system logs were collected to capture participants' interactions with the educational agent during the study sessions. These data sources included metrics such as time spent on tasks, prompt usage frequency, and action sequences. This information helped analyze user engagement, behavior patterns, and how participants navigated the system and utilized feedback to improve their CPS processes. For analysis, sequence analysis and descriptive statistics were applied to the screen recordings and system logs to understand user behavior and engagement patterns. Metrics such as time spent on tasks,

frequency of prompt usage, and interaction paths were examined to assess how participants used the agent's features (see Table 5). The analysis aimed to identify common strategies, variations in system use, and how these factors contributed to task outcomes.

6.4.4 Interview Data. In-depth interviews were conducted after participants completed both conditions to gather qualitative feedback on their experiences with the educational agent. The interviews explored participants' thought processes, the impact of system guidance, the challenges faced, and their overall preferences between the two conditions. The interviews were recorded, transcribed, and analyzed to identify themes and areas for potential system improvement. Thematic analysis was employed to analyze the qualitative data from the interviews [57, 74]. The coding process was iterative, focusing on refining themes that captured participants' experiences, perceived value, and suggestions for improvement. The results provided a comprehensive understanding of user experiences and were used to inform future iterations of the educational agent system (see supplementary materials for the Interview Protocol Outline). By combining these diverse measurement and analysis methods, our study provided a comprehensive assessment of Mentigo's effectiveness and usability in mentoring middle school students' CPS tasks. This mixed-methods approach enabled a thorough understanding of both the measurable outcomes and the subjective experiences of participants.

6.5 Expert Review and Feedback

To further validate the findings and gather insights into the system's effectiveness, we conducted an expert review and feedback session. We recruited five experts, including experienced secondary school teachers, a specialist in educational research, and a mentor with extensive experience in guiding middle school students. These experts reviewed the data generated from the study, such as students' learning reports, task outcomes, and dialogue records with the two systems. The experts evaluated the system's educational performance, usability, and impact on students' CPS skills. Following their review, they participated in a semi-structured interview to provide detailed feedback on how the system supported student engagement and learning outcomes. They highlighted the system's strengths and identified areas for improvement, offering practical suggestions based on their experience with middle school students. The insights from these expert sessions were crucial for refining our analysis of the two system performances. The feedback also provided actionable guidance for future development to enhance the

system's support for CPS and improve student learning experiences in real-world educational settings.

7 Findings

7.1 Evaluation of System Usability and Effectiveness

The Mentigo system is a data-driven intelligent agent, which is built on a robust foundation of over 80 hours of dialogue data collected from real CPS educational settings. It guides students through a CPS task using a six-stage process. At each stage, the system sets clear task requirements and progression standards, using corresponding guidance strategies and generative AI to ensure a personalized learning experience that adapts to each student's state and stages. Generally, students spent more time on tasks ($M = 43.7$ mins vs. 24.7 mins) and had more conversation rounds ($M = 24.0$ vs. 13.2) with Mentigo compared to the Baseline system, reflecting a more thorough and reflective learning process.

The Mentigo system demonstrated a high level of usability, with a System Usability Scale (SUS) score of 88.54, comparable to the Baseline system's score of 89.79. While both systems are highly usable, further analysis reveals significant differences in key performance dimensions that highlight Mentigo's strengths in guiding students through CPS tasks (shown in Table 6). The system excels in providing clarity of goals ($p = 0.051$) and delivering comprehensive guided strategies ($p = 0.032$), which enhances student engagement and promotes deeper learning. For instance, P5 said, *"The system A (Mentigo) actively guides you to complete the tasks, providing a structured and purposeful approach that helps us focus on our goals."* Students can follow its model to explore and learn specific knowledge, although it still requires students to do information integration and research, which encourages deeper engagement and learning. Mentigo outperformed the Baseline system in knowledge and skills acquisition ($p = 0.019$), effectively helping students learn to articulate issues and gain CPS knowledge. This aligns with the qualitative feedback where students expressed high appreciation for Mentigo's clarity in outlining and guiding tasks at each stage. P1 mentioned, *"System A (Mentigo) is more like a project mentor. It provides more guidance and leads you step-by-step, almost like a teacher guiding you through the process."*

Table 6: Comparison of Condition System and Baseline System Across Different Dimensions

Dimension	Mentigo	Baseline	Value	p
Helpfulness	$M=8.83$	$M=8.75$	$t=0.17$	0.867
Goal Clarity	$M=9.17$	$M=7.42$	$t=2.07$	0.051 [^]
Guided Strategy	$M=9.17$	$M=8.17$	$t=2.28$	0.032*
Personalization	$M=9.17$	$M=8.83$	$t=0.98$	0.339
Knowledge and Skills Acquisition	$Mdn=9$	$Mdn=7.5$	$U=112$	0.019*
Emotional Support	$M=8.33$	$M=7.42$	$t=1.18$	0.25

Mentigo's core strength lies in its ability to dynamically identify and respond to over 23 distinct student states, such as "unclear task," "overly divergent thinking," and "surface-level thinking" (see Table 3 for details). The analysis of student state frequencies shows that states like ID 1 ("Silent"), ID 2 ("Goal Unclear"), and ID 23 ("Normal Progress/Active Questioning") are most frequently addressed,

indicating these are common areas where students need support. Moderately frequent states, such as ID 10 ("Shallow Thinking"), ID 7 ("Limited Thinking"), and ID 13 ("Inadequate Information Integration Skills"), suggest that challenges in critical and creative thinking are also significant but less prevalent. Less common states, including ID 4 ("Goal Prioritization Confusion"), ID 17 ("Decision Uncertainty"), ID 18 ("Insufficient Intrinsic Motivation"), and ID 19 ("Lack of Confidence"), highlight Mentigo's capacity to provide targeted support for specific, less frequent student needs. This varied distribution of student states reflects Mentigo's adaptability and effectiveness in delivering personalized educational experiences tailored to diverse learning needs.

Similarly, the distribution of guided strategy IDs (see Table 2 for details) shows that strategies like ID 1 ("Task Follow-up and Allocation"), ID 15 ("Resource Guidance"), and ID 18 ("Encouragement and Motivation") are frequently employed, highlighting Mentigo's focus on these methods to support students effectively. Other strategies, such as ID 8 ("Probing Guidance"), ID 9 ("System Thinking Training"), and ID 13 ("Resource Guidance"), are also used often, indicating their targeted application for specific needs. Less frequent strategies, like ID 5 ("Creative Filtering and Focusing"), ID 11 ("Deep Thinking and Reflection"), and ID 19 ("Emotional Counseling"), demonstrate the system's adaptability in addressing unique challenges. This varied use of instructional strategies underscores Mentigo's ability to provide personalized and flexible educational experiences tailored to diverse learning contexts.

7.2 Student CPS Knowledge Improvement and Performance

Based on the analysis of learning outcomes, a Mann-Whitney U test was conducted to compare the pre-test results between the Mentigo and Baseline groups (as shown in Table 7). The results suggested that there were no significant differences between the two groups in their pre-test scores ($p = 0.51$). This indicates that both groups started with similar levels of knowledge before the intervention.

For the Mentigo group, there was a notable improvement from the pre-test to the post-test ($p=0.006$). The post-test scores ($M = 6.67$, $SD = 2.19$) were higher than the pre-test scores ($M = 4.33$, $SD = 2.23$), with a large effect size (Cohen's $d = 0.944$), suggesting substantial learning gains. In contrast, the Baseline group showed similar median scores between the pre-test ($M = 4.92$, $SD = 2.61$) and post-test ($M = 5$, $SD = 1.91$), though the effect size (Cohen's $d = 0.036$) indicates a medium difference, implying less change. When comparing the post-test scores between the Mentigo and Baseline groups, the effect size (Cohen's $d = 0.765$) suggests a large difference favoring the Mentigo group, and the Mann-Whitney U test results indicated that this difference was statistically significant ($p = 0.019$). P3 stated, *"I prefer the first system, Mentigo, because the baseline system often just tells you what to do directly, while Mentigo tells you the steps of thinking and guides you to think step-by-step."* This feedback highlights how Mentigo facilitates deeper cognitive engagement by guiding users through a structured thinking process, which aligns with the observed improvement in knowledge levels. On the other hand, P3, P8, and P9 mentioned that *"while the Baseline System is quicker to use and feels more like a tool that provides reference answers, it doesn't give a strong sense of accomplishment."*

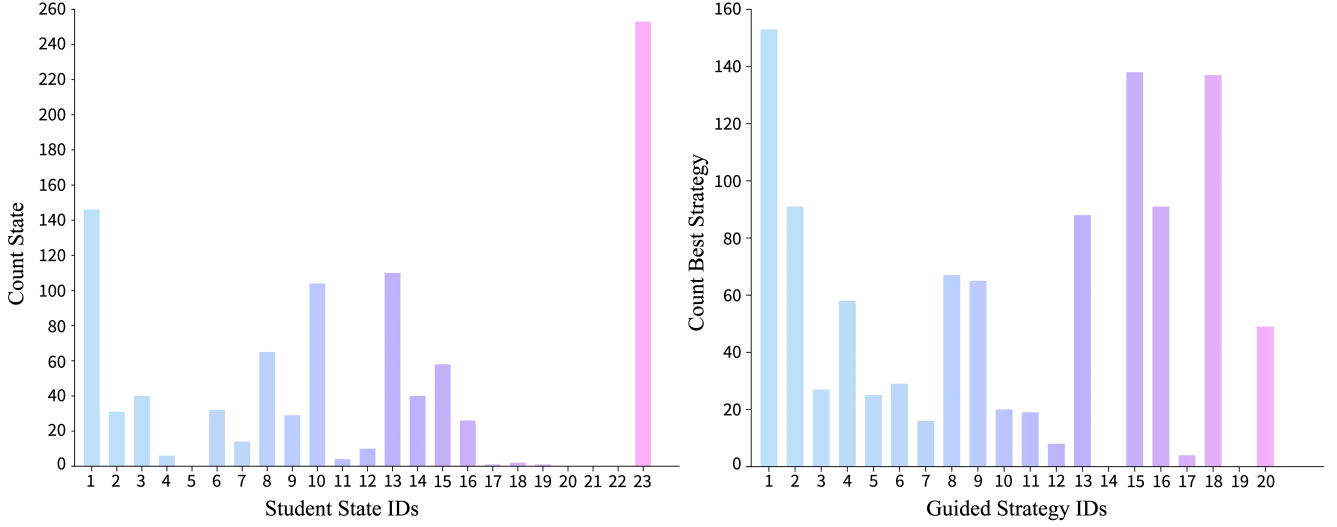


Figure 5: Distribution of Student States and Guided Strategies Frequencies

They described the experience as using a tool that gives quick solutions rather than fostering deep understanding. In contrast, *System A (Mentigo) helps us learn more methods and strategies and offers more guidance for critical thinking, leading to a richer learning experience and greater gains in knowledge and skills.* P12 added.

Table 7: Pre-/post-test Results (mean and standardized deviation) of Mentigo and Baseline

	Mentigo	Baseline	Effect size	<i>p</i>
Pre-test	4.33(2.23)	4.92(2.61)	0.19	0.51
Post-test	6.67(2.19)	5(1.91)	0.77*	0.019*
Effect size	0.944**	0.036		
p-value	0.006**	0.9		

Besides, we also evaluated the students' learning reports completed using the two systems across four dimensions. An independent samples t-test revealed a significant difference between the scores for Mentigo ($M = 13.43$, $SD = 3.17$) and the baseline system ($M = 10.74$, $SD = 2.72$), with $p = 0.016$. This result suggests that students' performance, as reflected in their learning reports, was significantly higher when using Mentigo compared to the baseline system. This is consistent with what P12 and P10 observed:

"By using Mentigo, I learned more methods and strategies, and I gained more from it. It pushes me step-by-step to complete all the tasks, and the ideas provided are based on the questions I raised, which often gives me a moment of clarity and leads me to think more comprehensively."

These comments suggest that Mentigo's tailored guidance and reflective approach lead to deeper understanding and more significant learning gains compared to the baseline system. P11 further added, "*Mentigo gives many prompts that allow me to think for myself, which makes me more creative because it focuses more on my own*

answers and guides me to write step-by-step." This insight indicates that Mentigo encourages more creative and independent problem-solving, enhancing the overall learning experience.

Overall, these results indicate that while both systems significantly impacted students' knowledge and performance, Mentigo provided a more structured and comprehensive learning experience that led to greater improvement in CPS knowledge and skills. The combination of quantitative data and qualitative insights suggests that Mentigo's approach to guiding students and encouraging deep thinking was more effective in enhancing their overall learning outcomes.

7.3 Student Engagement Across Mentigo and Baseline Systems

To analyze student engagement across the Mentigo and Baseline systems, we compared data on student-system interactions, including the number of dialogue turns and usage time. Additionally, we encoded the content of student-system dialogues to assess different aspects of engagement. Due to time constraints, two students did not complete the tasks in the last two stages of the experiment using our system, these incomplete data were excluded from the final analysis. While we measured task load across various dimensions and found no significant differences between the Mentigo and Baseline systems, our system did not increase cognitive load for students, and overall student engagement remained high. The comparison between the Mentigo and Baseline systems across different dimensions shows significant differences in several areas of student engagement: behavioral, cognitive, and emotional.

Behavioral Engagement. At the behavioral level, we assessed students' task duration, the number of conversation rounds, and the word count of student speech while using both systems (as shown in Table 8). The results showed that students had significantly longer task durations ($p = 0.005$) and more conversation rounds (p

Table 8: Comparison of Mentigo and Baseline System Across Different Dimensions

Categories	Variable	Mentigo	Baseline	Value	<i>p</i>
Conversation Metrics	Task Duration (min)	M = 43.7	M = 24.3	t = 3.19	0.005**
	Conversation Rounds	Mdn = 24.5	Mdn = 12.5	U=82.5	0.015*
	Word Count	Mdn = 1438.5	Mdn = 290	U=83	0.014*
Speech Type	Positive	M = 15.9	M = 7.2	t = 3.29	0.0041**
	Neutral	Mdn = 5.5	Mdn = 3.5	U=65.5	0.254
	Negative	M = 2.5	M = 2.7	t = -0.15	0.8812
Cognitive Level	Memory	M = 1.9	M = 1.5	t = 0.58	0.5717
	Understanding	M=2.4	M=2.1	t=0.3	0.7692
	Application	M = 5.1	M = 4.8	t = 0.13	0.9017
	Analysis	M = 4.3	M = 1.7	t = 2.16	0.0444*
	Evaluation	M = 3	M = 0.9	t = 2.64	0.0166*
	Creation	Mdn = 7.5	Mdn = 2	U=81.5	0.0180*

= 0.015) when using Mentigo. The word count of student speech was also significantly higher in the Mentigo group compared to the baseline group ($p = 0.014$). This observation aligns with our experimental findings, where students tended to think more deeply about the CPS questions while using Mentigo. The system's guided prompts encouraged them to complete each task step-by-step and to engage more proactively. This suggests that the structured guidance in Mentigo fosters higher levels of engagement and encourages students to think and participate more actively.

Cognitive Engagement. For cognitive engagement, we coded the content of students' speech according to type (Positive, Neutral, Negative) and cognitive level, based on Bloom's Taxonomy (six levels). Significant differences were found in the higher-order cognitive levels-specifically, "Analysis," "Evaluation," and "Creation"-with Mentigo outperforming the baseline system ($p = 0.044$, $p = 0.0166$, and $p = 0.0180$, respectively). This highlights how Mentigo, acting more like a mentor, encourages deeper cognitive engagement and supports the development of CPS skills through guided exploration and reflection. This highlights how Mentigo, acting more like a mentor, encourages deeper cognitive engagement and supports the development of CPS skills through guided exploration and reflection. P11 mentioned,

"When using the baseline system, it's easy to lose track and not know what to do next. But Mentigo has a more structured flow, making it less likely to go off-topic, and the guidance provided is very comprehensive, which leads to more complete thinking."

This statement underscores Mentigo's ability to provide a structured and comprehensive framework that helps students maintain focus and think more broadly, contrasting with the more direct answer-providing nature of the baseline system.

Emotional Engagement. In terms of emotional engagement, although quantitative results showed no statistically significant differences in emotional support between the two systems, the interview data provided further insights. Many students felt that Mentigo was more like conversing with a human, using simpler language, emojis, and interactive encouragements. For instance, P7, P8, and P9 noted that "Mentigo feels more like a peer, providing encouraging messages and engaging in more interactive dialogues. In contrast, the baseline system feels more like a reserved friend who responds when asked but

doesn't initiate conversations." This suggests that Mentigo provides a more supportive and engaging emotional environment, particularly suitable for middle school students, whereas the baseline system seems more like a hyper-rational expert that lacks warmth and adaptiveness in communication.

These findings indicate that Mentigo provides a more comprehensive and effective environment for student engagement across behavioral, cognitive, and emotional dimensions. It promotes active participation, higher-order thinking, and emotional connection, making it more suitable for fostering CPS skills and creativity among middle school students. The baseline system, while functional, appears to focus more on information retrieval and direct answers, with less emphasis on interactive and personalized learning experiences.

7.4 Expert's Review and Feedback

We conducted interviews with five experts to gather insights into students' cognitive changes, AI's guidance strategies, and interactions with different systems (Mentigo vs. Baseline). The experts reviewed dialogue data between students and the systems, along with learning report data, and provided comprehensive feedback from an educational perspective. Experts noted significant differences in how students interacted with the Mentigo system compared to the Baseline system. E1 observed, "It often seemed that students treated the baseline system more like a tool to command and directly get answers rather than engaging in deep thinking or dialogue." This suggests that the structured guidance in Mentigo fosters higher levels of engagement and encourages students to think and participate more actively. When reviewing the dialogues of P1 using Mentigo and the baseline system, E3 highlighted that as the conversation progresses, the students show "more comprehensive expressions and transfer previously mentioned content into new contexts. In contrast, the baseline system felt more like an information retrieval tool with limited cognitive advancement. E1 added, "The baseline system is more like a didactic expert who provides answers but does not engage with students at a deeper level." This difference highlights how Mentigo, acting more like a mentor, encourages deeper cognitive engagement and supports the development of CPS skills through guided exploration and reflection.

This observation was echoed by E2, who elaborated, "Mentigo offers more patient encouragement, like a cheerful artist friend who

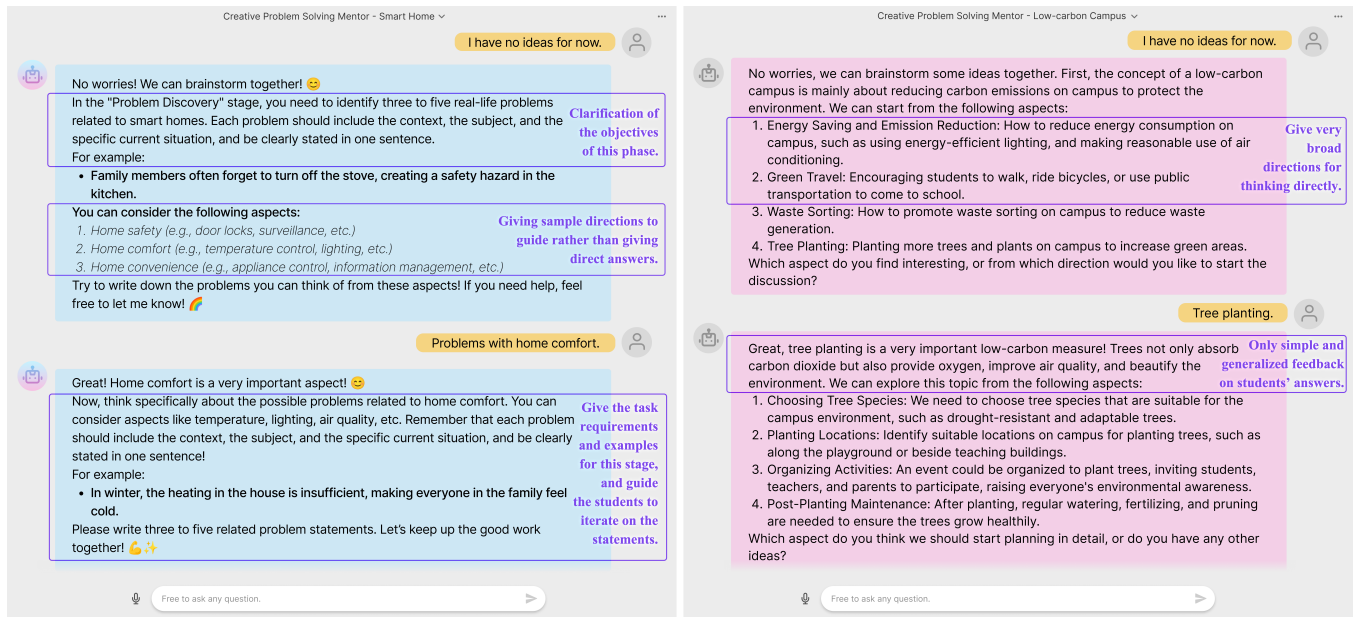


Figure 6: Illustration of Student P1's Dialogues with Mentigo (Left) and the Baseline System (Right) at the Same Stage of Different Tasks

is more expressive and proactive, even using emojis to ask questions." This supports the feedback from students who noted that Mentigo provided comprehensive guidance and multiple perspectives for thinking. E5 mentioned, "The guidance was simple yet offered various perspectives and space for thinking," suggesting that the system's ability to guide through structured steps supports its quantitative superiority in fostering cognitive engagement. In contrast, E4 commented that the Baseline System's approach often felt like a "question-and-answer session," lacking the same depth of guidance. Experts praised Mentigo's ability to provide structured guidance and maintain a comprehensive framework that helps students remain focused and think more deeply.

E5's feedback supports this observation: When P1 interacted with both systems at the beginning of the tasks, the differences in their guidance became evident. System A (Mentigo) provides clear direction at the start of each stage, explaining the task objectives and guiding students to discover more problems from multiple perspectives based on their input. It also prompts students to clarify their problem statements as required. Meanwhile, when P1 interacted with System B, it generated multiple answers based on the provided keywords for the student to choose from. This often led to students quickly adopting the AI-generated answers and ending the conversation prematurely. An excerpt of these interactions is shown in Fig. 6. This highlights Mentigo's role in scaffolding the learning process and helping students build their CPS skills step-by-step.

Experts also highlighted the contrast between the two systems' approaches: while Mentigo was described as a "mentor," the Baseline system was perceived as a "knowledge authority" that simply provides information. E3 pointed out that "System B (Baseline) felt like an over-rational tool; it lacked the 'human touch,' making it

difficult for students to relate and engage deeply." This difference impacted students' engagement levels, with Mentigo being more effective in fostering both cognitive and emotional engagement. Besides, as for the emotional support, E1, E3, and E4 stated that Mentigo was more like conversing with a human, using simpler language, emojis, and interactive encouragements. For instance, E4 noted that "Mentigo feels more like a peer, providing encouraging messages and engaging in more interactive dialogues." In contrast, the Baseline system feels more like a reserved friend who responds when asked but doesn't initiate conversations. This suggests that Mentigo provides a more supportive and engaging emotional environment, particularly suitable for middle school students, whereas the Baseline system seems more like a hyper-rational expert that lacks warmth and adaptiveness in communication. Besides, expert feedback suggests a significant opportunity to develop a teacher interface for real-world applications. E2, E3, and E5 both highlight Mentigo offers valuable insights into students' thinking processes and progress, which could be further enhanced by making these processes visible to educators in real-time. E2 stated,

I think a teacher interface could allow educators to monitor students' states, visualize their thought processes, and provide formative assessments based on real-time data. Such a feature would enable teachers to intervene more effectively, offer targeted support, and refine AI-driven guidance strategies, creating a more integrated and dynamic educational environment.

Overall, the experts' feedback suggests that the Mentigo system, with its structured and personalized guidance, offers a more effective and engaging environment for students compared to the Baseline system. It promotes cognitive development, encourages deeper engagement, and creates a supportive emotional atmosphere

that is crucial for fostering CPS skills. The insights gathered from the experts provide valuable directions for further refining the AI's guidance strategies to better support diverse student needs and learning styles.

8 Discussion

In this section, we reflect on Mentigo's design and evaluation, highlighting its contributions to educational AI, implications for student learning, and future research directions.

8.1 Mentoring Student with Mentigo

Effective mentoring is critical for fostering CPS skills, engagement, and learning in CPS tasks. Our study explored how AI-driven mentorship through Mentigo can enhance these outcomes by offering personalized, adaptive guidance. The findings reveal that Mentigo supports behavioral, emotional, and cognitive engagement by encouraging reflective dialogue and providing tailored support (see Section 7.3), consistent with prior research on creative process mentoring [46, 83]. While many AI-based educational systems emphasize delivering direct solutions [21, 69], Mentigo focuses on interactive mentoring that promotes deep learning and creative thinking. Its design represents the first implementation of a student-state-driven adaptive mentoring framework in a real-time learning environment. By continuously adjusting its guidance strategies based on learners' evolving cognitive, emotional, and behavioral states, Mentigo transforms the mentoring process from reactive responses to proactive collaboration. Compared to prior systems that rely on static prompts or scripted tutoring environments [4, 69], Mentigo dynamically selects mentoring strategies, adapting to unique learning contexts. This ability to respond contextually and adaptively highlights its potential to provide scalable, feasible mentoring solutions for developing CPS skills in educational environments. Our evaluation results affirm that Mentigo can serve as a robust, flexible, and impactful tool for supporting student learning and engagement.

The system's effectiveness is rooted in the development of a curated dataset of 2,362 real-world student-agent interaction sequences, enabling a dynamic mentoring strategy model that goes beyond simple task completion. This model leverages real-time learner signals to adapt its strategies, enabling proactive, context-aware interventions. Unlike traditional rule-based systems prone to hallucinated feedback [25, 31, 97], Mentigo minimizes inaccuracies by grounding its responses in authentic, context-sensitive student interactions. Previous systems relying on scripted learning environments often lack adaptability and ecological validity [76, 81]. In contrast, Mentigo's adaptive model supports nuanced learning contexts, offering more precise, personalized, and meaningful guidance while ensuring scalability and feasibility in real-time learning environments.

Mentigo represents the first AI-driven adaptive mentoring system for dynamic CPS tasks, advancing beyond rule-based and scripted educational systems. By combining real-time monitoring with adaptive strategy selection, it moves past static prompts to deliver personalized, context-aware support. Its ability to learn from real-world data opens new possibilities for scalable AI mentoring systems, paving the way for advancements in intelligent

tutoring, personalized learning platforms, and adaptive educational technologies.

8.2 Challenges of implementing in the educational settings for middle school students

Implementing AI-driven mentorship systems like Mentigo in educational settings offers promising opportunities but also presents unique challenges, particularly when targeting middle school students [34, 89]. This age group undergoes rapid cognitive and emotional development, making it a critical stage for fostering CPS skills through CPS tasks [37, 94]. Unlike adult learners, who have been the primary focus of prior research [70, 112], middle school students exhibit diverse cognitive readiness, emotional sensitivity, and attention spans, complicating the design of standardized mentoring approaches. Their expectations toward AI systems also influence how they engage with such technologies [25]. Designing an AI mentor for this demographic requires balancing structured guidance with open-ended exploration, keeping students focused while fostering CPS skills. Mentigo addresses this through dynamic scaffolding, adjusting its feedback strategies based on real-time interactions. However, ensuring sustained engagement, resolving confusion, and preventing disengagement remain key challenges requiring more advanced real-time monitoring and adaptive guidance.

Scaling Mentigo from a one-on-one environment to real-world educational settings introduces additional complexities. In classroom environments, challenges include facilitating group dynamics, supporting peer collaboration, and aligning AI-driven guidance with teacher-led learning objectives [25, 67]. For example, supporting collaborative CPS tasks requires Mentigo to mediate group interactions, ensure equitable participation, and adapt its mentoring strategies to accommodate both individual and group needs. Teachers play a vital role in managing classroom dynamics, integrating AI systems into curricula, and providing human oversight where AI may fall short [20, 30, 95]. Successful implementation thus requires not only technological improvements but also collaboration between AI developers and educators to align system capabilities with pedagogical goals. These multi-layered dynamics are often underexplored in prior research, underscoring the need for systems like Mentigo to evolve into tools that effectively support both teacher-led and collaborative learning environments.

8.3 Design Implications for Building Agents for Foster Student CPS Skills

8.3.1 Structured Task Management and Guided Strategies. Mentigo enhances student learning outcomes by employing a structured task management system and a diverse set of guided strategies, addressing DG1 and DG2. By breaking CPS tasks into manageable stages, the system ensures clarity, helps students stay focused, and reduces cognitive overload [8, 79]. Explicit instructions and progress-tracking mechanisms offer a clear roadmap for problem-solving while maintaining engagement, aligning with prior research on task management in educational systems [16, 69, 102]. To support creative exploration alongside structured guidance, Mentigo integrates diverse prompts and strategies designed to stimulate critical

thinking and encourage open-ended problem-solving. These strategies are grounded in a dataset of 2,362 real-world student-agent interaction sequences, enabling the system to adapt its guidance contextually and effectively (see Section 5.2.2). This dataset forms the backbone of Mentigo’s ability to balance structure and flexibility, addressing both procedural compliance and creative exploration. Future systems could build on this by incorporating real-time monitoring tools and expanding comprehensive interaction datasets to adjust task difficulty and mentoring strategies dynamically, providing personalized, context-aware guidance at scale.

8.3.2 Understanding Student States and Matching Strategies Dynamically. Prior research on AI-driven educational systems has often relied on static assessment models with limited adaptability to students’ cognitive and emotional states [19, 58]. Mentigo addresses this limitation by leveraging real-time state monitoring to dynamically match mentoring strategies based on patterns observed in 2,362 real-world student-agent interactions, addressing DG3 and DG4. Similar to context-aware learning systems explored in prior studies [45, 88], Mentigo detects behavioral signals such as hesitation, incomplete responses, and task completion rates to adjust its guidance adaptively (see Section 5.2.3). This dynamic approach supports not only task completion but also higher-order thinking skills such as analysis, evaluation, and synthesis. The dataset-driven design further enabled the development of Standard Operating Procedures (SOPs) that guide Mentigo’s decision-making process, ensuring context-sensitive feedback tailored to diverse learning profiles (see Section 5.1). Expert evaluations emphasized the importance of these flexible SOPs in offering personalized, adaptive support. Future AI-driven systems could enhance this framework by incorporating multimodal inputs (e.g., voice, gestures) to improve the detection of student states, enabling more responsive and adaptive guidance across diverse learning contexts.

8.3.3 Fostering Emotional Engagement Through Empathy. Emotional engagement is essential for effective mentoring [40, 42], and Mentigo demonstrates a strong capacity to create a supportive, empathetic learning environment. Consistent with prior research, its empathetic interactions foster emotional well-being and motivation, offering a “warmer” mentoring experience compared to traditional systems [41, 60, 77]. To meet DG5, Mentigo’s guidance strategies, informed by a dataset of real-world interactions, enable the system to address student frustration or hesitation through empathetic language and contextual restatements, reinforcing a positive learning atmosphere. While quantitative results did not reveal significant differences in emotional support, qualitative feedback from experts indicated that Mentigo’s empathetic design principles enhanced students’ sense of support and motivation (see Section 7.4). Future systems could build on this by incorporating affective computing technologies, such as real-time emotion recognition and sentiment analysis, enabling even more personalized and emotionally sensitive interactions. These advancements could ensure that agents effectively foster emotional engagement while complementing cognitive and behavioral support strategies.

8.4 Limitation and Future Work

While Mentigo demonstrates strong potential to enhance student engagement in CPS tasks, several limitations remain. System stability issues, such as occasional response delays, may hinder the user experience. The study’s limited scale, involving a specific group of middle school students in controlled conditions, affects its generalizability to broader educational contexts. Additionally, adaptive guidance capabilities, particularly real-time task difficulty adjustment, require further refinement to address diverse student learning needs. Long-term scalability and potential biases in AI-driven mentoring were only partially explored, including how different student profiles might experience varying levels of support. Addressing these concerns is critical for ensuring fairness, effectiveness, and widespread adoption.

Future work should involve deploying Mentigo in real-world classrooms to evaluate its effectiveness in authentic learning environments. Expanding its capabilities for collaborative learning could enable more complex group-based CPS tasks, facilitating peer interactions. Teacher-AI integration, supported by an intuitive interface for monitoring student progress and delivering real-time interventions, could enhance system usability. Bias mitigation strategies, such as fairness-aware algorithms and longitudinal evaluations, should be explored to promote equitable learning experiences across diverse student populations. These efforts will support the development of scalable, adaptive, and inclusive AI-driven mentoring systems for educational contexts.

9 Conclusion

We presented Mentigo, an AI-driven educational agent designed to enhance student engagement through structured guidance, personalized feedback, and empathetic interactions in CPS tasks. Mentigo delivers adaptive support tailored to CPS tasks by leveraging LLMs and real-world student-agent interaction data. A within-subject experiment comparing Mentigo with a baseline system showed increased behavioral, cognitive, and emotional engagement, highlighting its potential to support meaningful learning experiences. These findings demonstrate how AI-driven mentorship can provide personalized, scalable guidance, contributing to research in HCI and educational technology while informing future AI-driven learning environments.

Acknowledgments

We would like to extend our profound gratitude to the Lab for Life-long Learning and the Future Laboratory at Tsinghua University, as well as the Ministry of Education Philosophy and Social Sciences Innovation Team (Tsinghua University National Major Engineering Design Collaborative Innovation Team) for their invaluable support. Additionally, this work was supported by the National Natural Science Foundation of China (Grant No. 62441219) and the Beijing Municipal Science and Technology Project (Grant No. Z231100010323005).

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